Rajshahi University
Department of Computer Science and Engineering
B.Sc. Engineering Part-IV, Odd Semester Examination-2024
Course Code: CSE 4191, Course Title: Blockchain
Session: 2020-2021, 2019-2020
Full Marks: 52.5, Time: 3 Hours

[Answer any Six (06) questions taking at least Three (03) from each section.]

Section-A

1. a) What is blockchain? Explain the characteristics of public, private and consortium blockchains. 4.00

   b) Compare the distinguishing features of centralized, decentralized and distributed systems. Hence, draw the advantages of blockchain technology as a decentralized system. 4.75

2. a) What is cryptography, and why is it considered essential in block chain systems? 2.00

   b) Describe the working principle of asymmetric encryption and explain its role in block chain security. 3.00

   c) How does public-key cryptography help to ensure safe and verified transactions within block chain networks? 3.75

3. a) What is a bitcoin address? Explain the address generation procedure with appropriate diagram. 3.50

   b) What is double spending problem? Explain how this problem is addressed and solved in bitcoin network. 3.25

   c) Differentiate the bitcoin digital wallet with bank account. 2.00

4. a) What are the different types of transaction standards in bitcoin? Explain how a Segwit transaction minimize transaction cost. 4.00

   b) Alice wants to send 1 BTC to her son, but she wants him to receive it on his 18th birthday, not before. She creates a transaction like this: 2.00

Field	Value
Input	Alice's UTXO
Output	1 BTC → Son's address
nLockTime	October 8, 2030 (Unix time = 1917638400)

   Explain how and when the transaction will be added into the blockchain.

   c) What is block height? Explain the uses of block height field with an example. 2.75

Section-B

5.	a)	What is smart contract? What are the challenges of implementing smart contract? Explain with examples.	4.00
	b)	What is oracle? What are the different types of oracles in blockchain? Explain how oracle data flow takes place upon request from a smart contract.	4.75
6.	a)	What is Ethereum, and in what ways does it differ from Bitcoin?	2.00
	b)	What were the main motivations behind the creation of Ethereum?	2.75
	c)	Explain the role of Ether (ETH) within the Ethereum network.	2.00
	d)	Discuss some key challenges faced by Ethereum, such as scalability and network congestion.	2.00
7.	a)	Define transaction in Ethereum. What are the component fields of a legacy type transaction? Discuss the uses of them.	4.00
	b)	A user submits a transaction to the Ethereum blockchain. The transaction is estimated to require 45,000 **gas**. To ensure that miners include the transaction in the next block, the user sets the gas price to 30 **gwei**. i. What is the total transaction cost in **gwei**? ii. If 1 **gwei** = 10^{-9} **ETH**, how much ETH does the user pay in total?	3.00
	c)	What initial conditions must be satisfied for an Ethereum transaction to be considered valid? Explain.	1.75
8.	a)	What is Hyperledger? What benefits does Hyperledger provide that make it suitable for enterprise applications? Discuss the limitations of it.	4.00
	b)	Define Hyperledger fabric. How does it differ with Hyperledger? Discuss the advantages and industry applications of Hyperledger fabric.	4.50

University of Rajshahi
Department of Computer Science and Engineering
B.Sc. Engg. Part-4 Odd Semester Examination-2023
Course: CSE4161 (Blockchain)

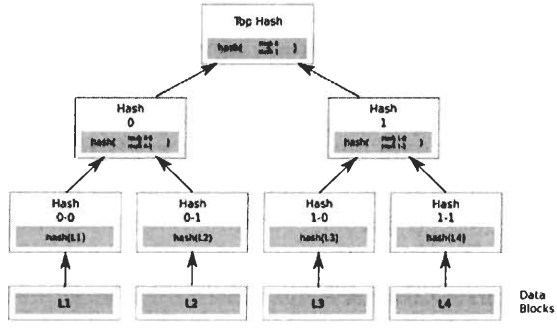
Full Marks: 52.5 Duration: 3 Hours

[Answer any SIX questions taking THREE from each section]

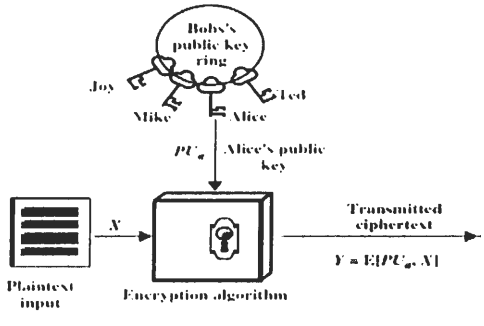
SECTION: A

1. a) What is cryptography, and why is it important in blockchain technology? 2
   b) How does asymmetric encryption work, and why is it used in blockchain? 3
   c) How does public-key cryptography ensure secure transactions in blockchain networks? 3.75

2. a) ***'Cryptographic hash function is the backbone of blockchain technology'***- Explain with the uses-cases of hashing in different phases of blockchain and cryptocurrency. 3
   b) Explain the ***avalanche effect*** in cryptographic hashing with an example. 2
   c) What is merkle tree? Explain what to do for the proof of L3's existence in the merkle tree. 3.75



3. a) Define symmetric and asymmetric key cryptographies. How do they differ with each other? 3
   b) What is digital signature? Explain the generation and verification procedures of digital signature with suitable examples for both of the cases. 5
   c) What will be needed to decrypt the ciphertext Y as viewed in the adjacent figure? 0.75



4. a) What is fault tolerance in consensus mechanism? How does CFT differ with BFT? 3
   b) What are the fundamental requirements for traditional consensus algorithms? Explain the working mechanism of the Paxos algorithm with a suitable diagram. 3.75
   c) A blockchain network is designed to be Byzantine fault-tolerant, meaning it can handle malicious or faulty nodes. If the network needs to tolerate 50 faulty nodes, determine the minimum number of nodes required to achieve consensus. 2

SECTION: B

5.	a)	What is Nakamoto consensus? Explain how the properties of Nakamoto consensus are mapped into the properties of traditional consensus.	3
	b)	What is the main drawback of CPU-bound PoW? How could it be solved using Memory-bound PoW?	2
	c)	Draw a diagram and explain how a generic PoS algorithm works to achieve consensus.	3
	d)	What is Sybil attack?	0.75
6.	a)	Define wallet import format (WIF) and mini private key format in bitcoin addressing.	2
	b)	Explain the step-by-step address generation procedure for addressing in bitcoin network. Use appropriate figure.	3
	c)	Illustrate the advantages and uses of vanity address and multisig address.	3
	d)	What is **P2SH**?	0.75
7.	a)	What is block height in bitcoin network? If a transaction has a lock time of block height 800000, what does it mean?	2
	b)	If a transaction uses two input UTXOs of 7 BTC and 9 BTC, and the recipient receives 10 BTC, how much Bitcoin will be returned as change?	2
	c)	What is 51% attack in bitcoin mining? Explain how you can mitigate 51% attack.	3
	d)	What steps should a miner take to evaluate whether Bitcoin mining is a viable and profitable investment?	1.75
8.	a)	How did Nick Szabo define smart contract? Explain the basic properties of smart contract.	4
	b)	Why are oracles necessary for smart contracts, and what limitations do they help overcome in blockchain systems?	2
	c)	Explain the standard mechanics of how an oracle interacts with a smart contract, including a generic data flow diagram to illustrate the process.	2.75